

Animesh

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TCS Research Fellow, IIT Kharagpur |  [Web](#) |  [Phone](#) |  [LinkedIn](#) |  [Github](#) |  [Google Scholar](#)

Education

Inidan Institute of Technology, Kharagpur (IIT-KGP) , PhD in Artificial Intelligence (CGPA: 8.4/10) Supervisor Prof. Plaban Kumar Bhowmick and Co-Supervisor Prof. Pralay Mitra	Aug 2021 – Ongoing
National Institute of Technology, Warangal (NIT-W) , PGD in Artificial Intelligence and Machine Learning (CGPA: 3.4/5)	Aug 2019 – Sept 2020
National Institute of Technology, Kurukshetra (NIT-KKR) , M.Tech in Control System (Electrical Engineering) (CGPA: 8.6/10)	July 2018 – July 2020
Pranveer Singh Institute of Technology, Kanpur (PSIT) , B.tech in Electrical and Electronics Engineering (CGPA: 7.4/10)	July 2012 – July 2016

PhD Thesis (Ongoing)

Thesis Title: *[“Equivariant Graph Neural Networks for Protein Interaction Modeling and Structure-Aware Molecular Design”](#)*

- Developed E(Q)AGNN-PPIS, an attention-augmented equivariant GNN framework that integrates complex spatial and geometric features for significant accuracy improvement.
- Designed a dynamic equivariant attention (GDEGAN) using adaptive attention kernels based on local feature statistics to accurately identify ligand-binding sites from 3D protein structure.

Research Experience

Research Internship , Ericsson Research, India Developed an efficient transfer learning architecture for multi-agent reinforcement learning (MARL) using graph neural networks for population-varying environments.	December 2025 - June 2026
AI Research Associate (ML, DL Instructor) , Spartificial, India Head of Artificial Intelligence team Took online lectures on machine learning and deep learning, with lab Helped establish the startup to educate in AI for space.	Oct 2021 – Aug 2022
Reviewer: I reviewed for the following conferences and Journals. Conferences: ICDM (PC Member)-26, IJCNN-27, 25, PC-Member-GCLR(AAAI-26), CRL Workshop@RLC-26, INDICON-24. Journals: ACM Transactions on Knowledge Discovery from Data (TKDD)-2025.	

Publications

1. GCT-MARL: Graph-Based Contrastive Transfer for Sample-Efficient Cooperative Multi-Agent Reinforcement Learning <i>Animesh</i> , Satheesh K. Perepu, Kaushik Dey Accepted, CRL Workshop@RLC, 2026	June 2026
2. Equivariant Graph Neural Networks for Protein Interaction Modeling and Structure-Aware Molecular Design <i>Animesh</i> IJCAI, DC (Short Paper), 2026	May 2026
3. PhysFlow: Physics-Informed Higher-Order Equivariant Flow Matching for Target-Aware Molecule Generation	May 2026

Animesh, Plaban Kumar Bhowmick, Pralay Mitra

[Under Review](#)

4. GDEGAN: Gaussian Dynamic Equivariant Graph Attention Network for Ligand Binding Site Prediction

March 2026

Animesh, Plaban Kumar Bhowmick, Pralay Mitra

[arXiv, 2026](#) [Under Review](#)

5. E(Q)AGNN-PPIS: Attention Enhanced Equivariant Graph Neural Network for Protein-Protein Interaction Site Prediction

June 2025

Animesh, Rishi Suvvada, Plaban Kumar Bhowmick, Pralay Mitra

[IEEE-Transaction on Signal and Information Processing over Networks](#)

6. Symbolic Hyperdimensional Vectors with Sparse Graph Convolutional Neural Networks

Jul 2022

Filip Cornell, Jussi Karlgren, **Animesh**, Šarūnas Girdzijauskas

[IJCNN, 2022](#)

7. Trajectory Tracking in a 3-DOF Robotic Manipulator using Sliding Mode Controller

June 2020

Animesh, Jyoti Ohri

[ICMICA, 2020](#)

Research Summer Schools

MLSS 2025: Attended the [MLSS 2025](#) Summer School and presented a research poster.

Location: Kraków, Poland Dates: July 1–6, 2025

ELLIS FoMo: Attended the [ELLIS Winter School on Foundation Models \(FoMo\) 2025](#).

Location: Tolhuistuin, Amsterdam Dates: March 18–21, 2025

Teaching Assistant

Course TA | AI60213: [FLLMs](#) | Foundation of Large Language Models, Autumn 2025-26, | IIT Kharagpur

Course TA | AI60007: [GMLFA](#) | Graph ML Foundations and Applications, Autumn 2022-25 | IIT Kharagpur

Course TA | AI42001: [MLFA](#) | Machine Learning Foundations and Applications, Spring 2021-22 | IIT Kharagpur

Course TA | AI60004: [BDP](#) | Big Data Processing, Spring 2022-25 | IIT Kharagpur

Summer School TA | Summer School: [MOML](#) | Maths of Machine Learning Summer School | IIT Kharagpur

Technical Skills

Language: Python, C++ | **Frameworks:** PyTorch, Pytorch-Geometric, DGL, TensorFlow, NetworkX, BioPython, e3nn | **others:** Git, Jupyter, LaTeX, NumPy, Pandas, Scikit-Learn etc.

Awards

- IJCAI-2026 Travel Grant.
- Received Prestigious Research Fellowship from TCS Research India, July-2023 [for 4 years]
- Awarded INSPIRE FELLOWSHIP by UP Government for Top 1 % in Intermediate board 2012

Certifications

- Deep Learning Specialization - 5 Courses | Neural Networks, Improving DNNs, Structuring a ML Project, CNNs, Sequence Models issued by [Coursera \(DeepLearning.AI\)](#)

Extra Curricular

Sports: Gym-training, Badminton, Cricket | **Others:** Reading books (biographies, lifestyle), Cooking, Writing (stories, poetry), Learning Flute, Listening to Podcasts (AI, Maths, Psychology etc.)